

Semigroups — Zero to Hero — Worksheet

James Mitchell¹, Reinis Cirpons², and Joseph Edwards¹

¹University of St Andrews

²Nantes Université, École Centrale Nantes, CNRS, Inria, LS2N, UMR 6004, France

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If you've never used GAP before you might want to look at:

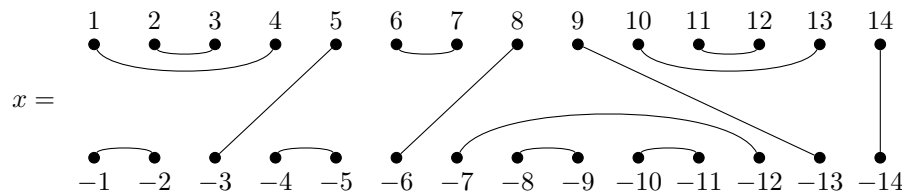
<https://carpentries-incubator.github.io/gap-lesson/>

before attempting any of the problems on this sheet.

This worksheet contains some exercises to be completed in GAP with the use of the **Semigroups** package as part of the GAP workshop session of NBSAN 40. The problems have been written in a deliberately terse way to encourage attendees to experiment with the package and explore its associated manual: https://semigroups.github.io/Semigroups/doc/chap0_mj.html.

As well as inspecting the package manual, please also feel free to ask questions to James, Reinis or Joe!

1. Let J_n denote the Jones monoid of degree n , and



- (a) How many elements does J_{14} contain?
 - (b) How many idempotent elements does J_{14} contain?
 - (c) How many idempotent elements does the Green's $\mathcal{D}$ -class of x in J_{14} contain?
 - (d) How many idempotent elements with 7 in the domain does the Green's $\mathcal{D}$ -class of x in J_{14} contain?
2. Recall that the commuting graph of a semigroup S is the graph with nodes the elements of S and an edge (x, y) if $xy = yx$ holds. Let T_n be the full transformation monoid on $n \leq 5$ points. Show that the clique numbers of the commuting graph of T_n are 2^{n-1} .
 3. Let S be the semigroup defined by the presentation

$$\langle a, b, c, d, e, f, g \mid abcd = aaeaa, ef = dg \rangle.$$

- (a) Show that S is infinite.
 - (b) Create a homomorphism from the free semigroup $\{a, b, c, d, e, f, g\}^+$ to S .
 - (c) Partition the first 1000 elements of $\{a, b, c, d, e, f, g\}^+$ so that words belong to the same part if and only if they represent the same element of S .
4. Is the semigroup defined by the presentation

$$\langle x_2, \dots, x_n \mid x_i^2 = (x_i x_j)^3 = (x_i x_j x_k)^4 = 1 \quad i, j, k \text{ distinct} \rangle.$$

the symmetric group?

5. Determine which of the relations in the presentation are redundant and which are not:

$$\langle a, b \mid a^4 = 1, b^2 = b, ba^3ba = a^2(ab)^2, (ba^2)^2 = (a^2b)^2, (ba)^2a^2 = aba^3b, a(ab)^4 = (ab)^4 \rangle$$

What is the minimal set of the relations in this presentation that define the same monoid?

6. Let S be the Catalan monoid of degree 3.
- (a) Draw the egg-box diagram of S .
 - (b) Draw the left and right Cayley graphs of S .
 - (c) Show that S has two non-trivial non-universal non-Rees congruences.
 - (d) Show that the lattice of left and right congruences of S are isomorphic.
 - (e) What are the maximal subsemigroups of S ? Show that none of the maximal subsemigroup is isomorphic to any of the others.
7. Let S be the dual of the full transformation monoid on 5 points. Find a transformation representation of S on 32 points.
- (a) Draw the egg-box diagram of S .
 - (b) Draw the left and right Cayley graphs of S .
 - (c) Show that S has two non-trivial non-universal non-Rees congruences.
 - (d) Show that the lattice of left and right congruences of S are isomorphic.